

Algebra: Evaluating Algebraic Expressions



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DIRECTIONS

Substitute the given value(s) for each variable, then simplify using the order of operations.

1. Evaluate when $x = 4$:

$$3x + 5$$

Answer: _____

2. Evaluate when $x = -2$:

$$2x - 7$$

Answer: _____

3. Evaluate when $a = 5$:

$$a^2 - 3a$$

Answer: _____

4. Evaluate when $n = -3$:

$$n^2 + 2n - 1$$

Answer: _____

5. Evaluate when $x=3, y=-5$:

$$2x - 3xy + 4y$$

Answer: _____

6. Evaluate when $x=2, y=-1$:

$$x^2 - 4xy + 3y^2$$

Answer: _____

7. Evaluate when $a=3, b=-2$:

$$a^2 + ab - b^2$$

Answer: _____

8. Evaluate when $p=2, q=-3$:

$$p^2 + 2pq - q$$

Answer: _____

9. Evaluate when $r=5, s=-2$:

$$2r - s^2 + rs$$

Answer: _____

10. Evaluate when $m=4, n=-1$:

$$3m^2 - 2mn + n^2$$

Answer: _____

Based on the Numberbender lesson "ALGEBRA: Evaluating Algebraic Expressions" • youtu.be/ODXWMgINUYg

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Answer Key & Solutions

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TEACHER NOTES

Items 1-4: single-variable substitution. Items 5-10: two-variable substitution with negative values — watch sign rules carefully.

1. Evaluate when $x = 4$: $3x + 5$

Answer: 17

$$3(4) + 5 = 12 + 5 = 17.$$

2. Evaluate when $x = -2$: $2x - 7$

Answer: -11

$$2(-2) - 7 = -4 - 7 = -11.$$

3. Evaluate when $a = 5$: $a^2 - 3a$

Answer: 10

$$(5)^2 - 3(5) = 25 - 15 = 10.$$

4. Evaluate when $n = -3$: $n^2 + 2n - 1$

Answer: 2

$$(-3)^2 + 2(-3) - 1 = 9 - 6 - 1 = 2.$$

5. Evaluate when $x=3, y=-5$: $2x - 3xy + 4y$

Answer: 31

$$2(3) - 3(3)(-5) + 4(-5) = 6 + 45 - 20 = 31.$$

6. Evaluate when $x=2, y=-1$: $x^2 - 4xy + 3y^2$

Answer: 15

$$(2)^2 - 4(2)(-1) + 3(-1)^2 = 4 + 8 + 3 = 15.$$

7. Evaluate when $a=3, b=-2$: $a^2 + ab - b^2$

Answer: -1

$$(3)^2 + (3)(-2) - (-2)^2 = 9 - 6 - 4 = -1.$$

8. Evaluate when $p=2, q=-3$: $p^2 + 2pq - q$

Answer: -5

$$(2)^2 + 2(2)(-3) - (-3) = 4 - 12 + 3 = -5.$$

9. Evaluate when $r=5, s=-2$: $2r - s^2 + rs$

Answer: -4

$$2(5) - (-2)^2 + (5)(-2) = 10 - 4 - 10 = -4.$$

10. Evaluate when $m=4, n=-1$: $3m^2 - 2mn + n^2$

Answer: 57

$$3(4)^2 - 2(4)(-1) + (-1)^2 = 48 + 8 + 1 = 57.$$

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